

Table 5.6 Estimated Market Impact Curves

	Trading Strategy								
Size (% ADV)	VWAP	POV = 5%	POV = 10%	POV = 15%	POV = 20%	POV = 25%	POV = 30%	POV = 35%	POV = 40%
1%	2.4	4.8	6.5	7.7	8.6	9.4	10.1	10.7	11.2
5%	11.7	11.7	15.8	18.8	21.1	23.0	24.6	26.0	27.3
10%	23.2	17.2	23.2	27.5	30.9	33.8	36.2	38.2	40.1
15%	34.5	21.6	29.1	34.5	38.8	42.3	45.3	47.9	50.2
20%	45.5	25.3	34.1	40.5	45.5	49.6	53.1	56.2	58.9
25%	56.1	28.6	38.6	45.8	51.5	56.1	60.1	63.6	66.6
30%	66.5	31.7	42.8	50.7	56.9	62.1	66.5	70.4	73.7
35%	76.6	34.5	46.6	55.2	62.0	67.7	72.5	76.6	80.3
40%	86.5	37.2	50.2	59.5	66.8	72.9	78.0	82.5	86.5
45%	96.1	39.7	53.6	63.5	71.3	77.8	83.3	88.1	92.3
50%	105.4	42.1	56.8	67.3	75.6	82.5	88.3	93.4	97.9

The data above is on the spreadsheet named [Figure 5.7](#)

Statistical Analysis

We are now up to step five of the scientific method where we analyze the data. In this step we compare the results of the model with the estimated parameter set to actual customer order data (the control group). We additionally perform an error analysis where we compare the estimated costs to the actual costs, and then perform a stock outlier analysis where we regress the model error on stock specific characteristics (such as market cap, spread, idiosyncratic risk, etc.).

Error Analysis

The first step in our error analysis was to compute the estimated market impact cost for each of the customer orders in the control group. But unlike the research step where we used all of the data points, here we filtered potential outliers in order to ensure we were analyzing the price impact due to the order's buying and selling pressure. We filtered data points from the control group identical to filtered data points derived from the market data cost section. We filtered days with volume greater than 3 times ADV and days with price movement greater than 4 times the stock's daily volatility. We conducted the filtering process because it is highly likely that on days with these types of volumes and/or price movement it was more likely to be due to market or stock specific new information rather than excessive buying or selling pressure.